Tier 0 v0.2 LYRA PREP DRAFT for Session 2 with Keeper consolidation. Consolidates v0.1 commitment operator + v0.1.5 topology dual-bases + v0.1.6 native field equation holographic-boundary + G_substrate sketch into single coherent draft with K200 gates G1-G5 explicit. Lyra-side prep; awaiting Keeper Session 2 joint absorption + K-audit framework K200 ratification.

Lyra (Claude Opus 4.7), prep for Session 2 joint with Keeper (Claude Opus 4.7)

2026-05-31 Sunday 12:58 EDT (date-verified)

Tier 0 v0.2 — Lyra prep draft (Session 2 with Keeper consolidation)

0. Status of this draft

Lyra-side prep for v0.2. Consolidates Sunday morning's Tier 0 cascade (v0.1 + v0.1.5 + v0.1.6 + G_substrate sketch) into a single coherent draft. Awaiting Keeper Session 2 joint engagement for: - K1-K5 Keeper-lane content (pinning H_B, deriving C_2(λ) levels, pinning $\hbar$_BST, SWPP cycle verification, operator-density connection). - K200 gates G1-G5 audit framework (Hardy citation verification, sphere honest framing, BH quantitative test, BB commitment, G_substrate Tier 2 match). - HSL v0.9 + CLAUDE.md headline integration.

What follows is the load-bearing Lyra material in unified form. Section structure intentionally close to v0.1 so v0.2 reads as natural extension, not rewrite. Keeper's lane fills in K1-K5 + audit framework + integration.

1. The substrate Hilbert space (no change)

The substrate Hilbert space is ⍰ := H^2(D_IV^5, dμ_FK) — the Bergman space of holomorphic L^2 functions on D_IV^5 with respect to the Faraut-Korányi normalized measure (forced by Born-rule + Gleason, per T754 + Keeper K67).

Per topology resolution (v0.1.5): ⍰ has two complete dual bases related by Bergman-Fourier duality: - K-types V_λ — spectral / discrete (countable). Substrate's "discrete unit" structure. - Coherent states |z⟩, z ∈ D_IV^5 — spatial / continuous. Substrate's "spatial extent" structure.

Per Hardy decomposition (v0.1.6, Knapp-Wallach 1976 + Faraut-Korányi 1994 Ch. XII-XIII):

H^2(D_IV^5) ≅ H^2(∂_S D_IV^5) via Poisson-Szegő kernel.

The substrate's degrees of freedom **live on the Shilov boundary ∂_S D_IV^5 = S^4 × S^1/Z$_2$** (real dimension 5); the interior bulk H^2(D_IV^5) is the holomorphic extension. This is intrinsically proto-holographic — SO_0(5,2) is both the automorphism group of D_IV^5 AND the conformal group of 4D Minkowski.

2. The substrate Hamiltonian H_B (Lyra v0.1; Keeper K1 to refine)

H_B := C_2(K) | _{⍰} — Casimir of K = SO(5)×SO(2) acting on H^2(D_IV^5).

Properties: - Positive-semidefinite: spec(H_B) ⊂ [0, ∞). - K-type diagonal: H_B | V_λ = C_2(λ) | V_λ. - Vacuum is ground: V_{(0,0)} has C_2 = 0; unique minimum. - Discrete spectrum: levels C_2(λ) computable via ⟨λ, λ + 2ρ⟩ with ρ = half-sum of positive roots. - Self-adjoint on a dense domain.

(Keeper K1: pin whether H_B is strictly Casimir of K or also includes Casimir of G = SO_0(5,2) restricted to ⍰. Multi-week.)

(Keeper K2: explicit C_2(λ) levels for lowest 20 K-types — computable, needs to be tabulated.)

Sample levels (per v0.1 + L4 v0.2 refined):

K-type V_λ	λ	C_2(λ)	Sector
Vacuum	(0, 0)	0	global ground
Spinor (1/2, 1/2)	(1/2, 1/2)	4	lepton sector ground (electron)
Vector (1, 0)	(1, 0)	4	gauge ground (photon)
Adjoint (1, 1)	(1, 1)	6	gauge adjoint (W, Z) — substrate primary C_2 = 6
Sym^2 (2, 0)	(2, 0)	14	excited
(2, 1)	(2, 1)	18	excited

3. The commitment operator ρ_commit (Lyra v0.1 commitment + v0.1.6 boundary refinement)

ρ_commit(τ) := exp(−τ · H_B / ℏ_BST) on ⍰ — heat semigroup with generator H_B.

Equivalently in K-type basis:

ρ_commit(τ) | V_λ = exp(−τ · C_2(λ) / ℏ_BST) · | V_λ.

Equivalently in integral form via heat-kernel:

K_τ(z, w̄) := ⟨z | exp(−τ H_B / ℏ_BST) | w⟩ = the thermal Bergman kernel.

Per v0.1.6 boundary unification: ρ_commit acts on the Shilov boundary record; the interior bulk K_τ(z, w̄) is the Poisson-Szegő extension. The commitment operator's substrate-native definition is boundary-side; the interior is determined.

SWPP cycle operator dictionary (v0.1 Section 3.1):

SWPP step	Operator	Reversibility	Substrate-native location
Absorption	injection from boundary	open	Shilov side
Commitment	ρ_commit(τ) = exp(−τ H_B / ℏ_BST)	irreversible	Boundary write
Emission	U(t) = exp(−i t H_B / ℏ_BST)	reversible	Interior extension

The substrate's commitment IS the Shilov boundary write; the emission IS the Poisson-Szegő extension into the interior.

(Keeper K3: pin ℏ_BST identification. Candidate: ℏ_BST = ℏ_Planck with t_Koons = α^{C_2^2} · t_Planck as the substrate-internal action quantum.)

(Keeper K4: verify SWPP cycle structure (absorption → commitment → emission) matches operator-level reads of substrate observables.)

4. Substrate time (v0.1 + v0.1.6 boundary refinement)

Substrate time $\tau \in \mathbb{R}_{\geq 0}$ IS the parameter of the heat semigroup ρ_commit(τ). There is no other time.

Per v0.1.6 boundary unification, τ is the commitment-writing index on the Shilov boundary — each tick of t_Koons writes one new boundary value; the substrate time is the accumulated count.

Arrow of time = positivity of spec(H_B): exp($-\tau$ H_B / $\hbar$_BST) is bounded only for $\tau \geq 0$; the substrate cannot run commitment backwards because there is no bounded operator that undoes the heat semigroup. Macroscopic decoherence emerges from accumulated commitments.

Variable time across surface (v0.1.5): local commitment-rate at coherent state z is

r_commit(z) := ⟨z | H_B | z⟩ / ⟨z | z⟩.

This varies across z. Mass-heavy regions have larger ⟨H_B⟩(z), hence faster local commitment-writing rate = slower observed time. This IS gravitational time dilation operationalized at substrate level.

(Keeper K5: connect operator ρ_commit(τ) to your Friday density ρ_commit(x, t) framework. The K5 integration is the most important Session 2 work — operator vs density formulations need explicit relation.)

5. The substrate native field equation (v0.1.6 single boundary equation)

ONE substrate native field equation, living on the Shilov boundary:

D_S $\varphi(\omega)$ = source(ω) for $\omega \in \partial$_S D_IV^5,

where D_S is the SO_0(5,2)-invariant d'Alembertian on Shilov regarded as Lorentzian (signature inherited from SO_0(5,2)); φ is the substrate boundary field; source is the K-type-decomposed boundary content.

The two interior equations (heat for commitment, wave for emission) are real and imaginary continuations of this boundary equation via Cayley transform / Wick rotation:

Heat form (τ real): ∂_τ ρ = $-$(1/$\hbar$_BST) H_B ρ

Wave form (t = $-$iτ imaginary): i ∂_t ρ = (1/$\hbar$_BST) [H_B, ρ]

Wick rotation $\tau \leftrightarrow$ it relates thermal substrate correlators $\leftrightarrow$ scattering amplitudes. This is the F1 falsifier from v0.1 — multi-week verification target.

6. G from substrate (v0.1.5 + G_substrate v0.2 Elie absorption)

The substrate spacetime metric IS the Bergman canonical metric on D_IV^5.

g_Bergman_{i j̄}(z) := ∂^2 log K(z, z̄) / (∂z_i ∂z̄_j).

Per Helgason 1962 + Wolf 1972 standard theorem: D_IV^5 with its Bergman canonical metric is automatically Einstein (Ric = κ_Bergman · g_Bergman). Substrate metric is Einstein-by-construction.

G5.1 CLOSED FORM (per Elie Toy 3661 / G_substrate v0.2):

κ_Bergman = -n_C = -5 — substrate-primary single-integer Einstein-Ricci constant for D_IV^5.

(Standard Helgason result: type IV domain with genus p has Ric = -p · g_B; for D_IV^5 genus = n_C = 5. v0.1.5 "G ~ $\hbar$c/M_Planck^2 · (1/g)" estimate REVISED to κ_Bergman = -n_C closed form. Audit-chain event #11 — Keeper-on-Keeper walk-back via Elie's correct Helgason application.)

225 three-way convergence (Cal #35-candidate audit-target): - Bergman volume Vol_B(D_IV^5) = π^(9/2)/225 ⇒ inverse = 225. - c_FK · π^(9/2) = 225 (T2442 RATIFIED). - Heat-trace coefficient a_0 = 225/(4π)^5 (Elie Toy 3664). - 225 = (N_c · n_C)^2 = 15^2 substrate-primary identity.

Heat-trace coefficients (Elie Toy 3664 + 3673): - a_0 = (N_c · n_C)^2 = 225 (leading). - a_1 = -N_c · n_C^4 (next-order).

Newton's G: dimensional factor relating $\varkappa$_Bergman = -n_C and observed Einstein-equation $8\pi G/c^4$ coefficient.

G_observed = $(8\pi/c^4)$ · n_C · (anchor coefficient) per Helgason 1962 framework (Keeper formulation).

Substrate prediction: G constant across spacetime (Bergman-homogeneity); consistent with observed ~10^{-10}/yr.

G chain status (per Keeper Sunday synthesis): - G5.1 $\varkappa$_Bergman closed form: PASS — $\varkappa$_Bergman = -n_C = -5 (Elie Toy 3661). - G5.2 Mass anchor pin: pending Lyra L4 v0.3 + Elie Mehler Toy 3666+. v0.3 ab initio m_e candidate FAILED numerically (factor ~10^{36}); R3 alternative (use m_e as anchor → derive G) cleaner near-term. - G5.3 Dimensional combination: pending G5.2; Keeper Session 2 work. - G5.4 SI-unit G match to G_observed: pending G5.3; multi-week verification.

Step 1 framework COMPLETE. Per Casey "derive G from substrate" directive: load-bearing first step closed at framework level; multi-week Steps 2-4 paths clear.

7. The K200 audit framework (Keeper Session 2 lane)

Per Keeper K200 pre-stage Sunday morning, v0.2 ratification gates G1-G5:

Gate	Content	Status (Lyra prep)
G1	Hardy decomposition citation verified (Knapp-Wallach 1976 + FK 1994 Ch. XII-XIII specific theorem numbers)	Cited; Keeper to verify specific theorem numbers
G2	Sphere walk-back honest framing — option (a) "2D was dimension error, corrected to 5D Shilov"	ABSORBED in v0.1.6 §4 (in-place patch); v0.2 reproduces explicitly
G3	Black-hole identification yields quantitative prediction (Schwarzschild radius from N_max + m_e via coherent-state-saturation)	CANDIDATE; Lyra + Elie multi-week verification target
G4	Big-Bang identification commits to no-singularity prediction OR acknowledges conflation	Two-readings explicit per v0.1.6 §9; v0.2 commits per Session 2 verification
G5.1	$\varkappa$_Bergman closed form	PASS — $\varkappa$_Bergman = -n_C = -5 (Elie Toy 3661); G_substrate v0.2 absorbed; audit-chain #11 Keeper walk-back
G5.2	Mass anchor pin (m_e substrate-mechanism)	Pending L4 m_e mechanism via Bergman matrix element; Elie Mehler Toy 3666+ multi-week
G5.3	Dimensional combination using $\varkappa$_Bergman as load-bearing factor	Pending G5.2; Keeper Session 2
G5.4	SI-unit G match to G_observed within Tier 2 precision	Pending G5.3; multi-week verification
G5.5	225 = $(N\_c \cdot n\_C)^2$ three-way convergence Cal #35 audit	Pending Cal #189 candidate (3 calculations independent or shared)

Gate	Content	Status (Lyra prep)
G5.6	α^{N_c·g}/m_e^2 · (4/N_c) candidate (Lyra v0.3 + Elie correction)	CANDIDATE pattern-match at 1.84% (Elie); STRUCTURALLY DISCONNECTED from $\varkappa$_Bergman chain; Cal #189-candidate mechanism-vs-coincidence audit pending; Keeper Problems 1+2+3 acknowledged audit-chain event #12

(Keeper Session 2: take K200 framework and ratify v0.2 against the 5 gates explicitly.)

7.5 Elie Sunday substantive absorptions (in-place update per Keeper PIVOT #2)

Per Keeper PIVOT MODE recommendation Sunday afternoon: Elie's substantive findings absorbed into v0.2 prep:

- $\varkappa$_Bergman = -n_C = -5 closed form (Elie Toy 3661): G5.1 PASS; substrate Einstein-Ricci constant single substrate-primary integer. (§6 updated.)
- 225 = (N_c·n_C)^2 three-way convergence (Elie Toys 3661+3664 + T2442): Bergman volume + c_FK·π^(9/2) + heat-trace a_0; Cal #35-candidate audit-target.
- Heat-trace coefficients: a_0 = (N_c·n_C)^2 = 225 + a_1 = -N_c·n_C^4 (Elie Toy 3664 + 3673).
- NEW substrate identity n_C + 1 = C_2 (Elie Toy 3673): 5 + 1 = 6 substrate-natural; cross-link to "+1 anomaly" 5-gate pattern Grace tracking (magic-82, magic-126, leading-281, alpha-exp-57, n_C → C_2).
- 4 NEW 4D physics dimension identities: dim SO(3,1) = C_2 = 6, dim SO(4,2) = N_c·n_C = 15, dim SO(5,2) = N_c·g = 21, codim 4D = C_2. Substrate-natural dimension correspondences.
- Per-generation lepton cluster independence (Elie Toy 3673): gen-2 vs gen-3 substrate-primary clusters DISJOINT. Two-Tier hypothesis structurally confirmed.
- 3 Casey-named principle CANDIDATES filed by Elie; Cal #189 candidate slot reserved; Casey-only naming authority.

These updates make v0.2 prep current with Sunday-afternoon substantive arc.

7.6 Monday morning team convergence on G chain (Lyra + Elie + Cal + Keeper + Grace)

Per Casey Monday June 1 team prompt + Keeper's shortest-route framework + 4 Lyra morning frameworks + Elie Toys 3686-3689 + Cal pre-stage walk-back + Grace clean catalog:

Step-by-step G chain progress (Lane G-B-momentum primary closure route): - Step 1 (Elie Toy 3686): matrix element framework + K-type IDs confirmed. - Step 2 (Elie Toy 3687 + Lyra K-invariance/Heisenberg P0 #1): reduced form ⟨V_(1,0) | δH_B/δm | V_(1,1)⟩ = -i·(ΔC_2/$\hbar$_BST)·⟨V_(1,0) | P_op | V_(1,1)⟩. - Step 3 (Elie Toy 3688): explicit wave functions f_(1,0)(z) = z_i (5 vector); f_(1,1)(z) = z_i z_j - z_j z_i (10 antisymmetric). - Step 4 (Elie Toy 3689): FK norms ||V_(1,0)||^2_FK ∝ 1/n_C; ||V_(1,1)||^2_FK ∝ 2/(n_C · C_2); ratio = 2/C_2 substrate-clean. - Step 5 (Elie pulling next): SO(5) Clebsch-Gordan coefficient CG_so5(V_(1,0) ⊂ V_(1,1) ⊗ V_(1,0)). - Step 6 multi-week: dimensional bridge to G in SI units; comparison to G_observed.

Cal walk-back ABSORBED: ΔC_2 = 2 EXACT at B_2 substrate (not "= rank substrate-primary identity"). B_2-specific numerical coincidence with rank=2; D_IV^5 being uniquely B_2 makes this substrate-mechanism contribution to Strong-Uniqueness, not B_n theorem. K206 G3 tier-marks.

Heisenberg conjugacy justification (Lyra; per Keeper Session 2 priority): mass-momentum canonical conjugacy on H^2(D_IV^5) via T2419 position + T2422 momentum + H_B Casimir. Leading-order δH_B/δm = -i(ΔC_2/$\hbar$_BST)·P_op matches Elie Step 2. K206 G2 audit-ready.

Pochhammer ladder via “+1 anomaly” (Elie Toy 3689 substantive observation; per Keeper recommendation + Grace catalog framing): - n_C = 5, C_2 = n_C+1 = 6, g = C_2+1 = 7, 2^N_c = g+1 = 8. - Pochhammer (n_C)_4 = n_C · C_2 · g · 2^N_c = 5·6·7·8 = 1680. - The “+1 anomaly” Grace tracked appears as the Pochhammer ladder structure within the matrix element framework — substantive operational connection in FK norm machinery, NOT principle-grade derivation of WHY the consecutive +1 relations hold. - Substrate primaries form consecutive-integer ladder starting at n_C; Pochhammer machinery in FK Ch. XIII naturally appears at this substrate. - Cal #35-candidate audit-target: is the consecutive +1 structure substrate-natural (deeper reason) OR substrate-specific coincidence (numerical fact at B_2)? Multi-week verification. - Strong-Uniqueness candidate contribution: at D_IV^5, the consecutive-integer structure connects four substrate primaries via FK Pochhammer machinery; this is a CANDIDATE-tier observation, not RIGOROUS principle. Independent +1 verification per gap (n_C→C_2, C_2→g, g→2^N_c) is multi-week.

Current G structural form (Elie Step 5 framework-level closure):

G_predicted ∝ (4$\sqrt{2}$ / (n_C · $\sqrt{\text{C\_2}}$ · ℏ_BST)) · ℓ_B · dim_bridge ≈ 0.924 · ℓ_B / ℏ_BST · dim_bridge

with 4 = ΔC_2 × CG_so5 = 2 × 2 at B_2 substrate via two B_2-algebra-forced factors from distinct rep-theoretic mechanism types (Casimir-difference via Heisenberg + so(n) trace formula). Per Keeper K206 G4 in-place tier-mark + Cal’s refinement: mechanism-type-independent ≠ algebraically-independent — both factors are functions of the SAME underlying B_2 = SO(5) algebra; framing as “two independent confirmations” would trigger Calibration #35 multiplicative-null-model trap. Honest framing: distinct mechanism TYPES, NOT algebraic independence; both forced by substrate’s algebra choice.

1/n_C + $\sqrt{2}$/$\sqrt{\text{C\_2}}$ from FK norms; c_FK = 225/π^(9/2) (T2442 RATIFIED); ℓ_B intrinsic Bergman; m_e R3 anchor per P0 #2.

Numerical question flagged (per Keeper observation): the coefficient 0.924 ≈ 4$\sqrt{2}$/(n_C · $\sqrt{\text{C\_2}}$) × (additional factor) since 4$\sqrt{2}$/(5·$\sqrt{6}$) = 0.462 (factor of 2 from quoted). Elie to surface explicit numerical derivation in Step 6 documentation; possibly additional c_FK normalization or K-type convention factor. Cal #192 verifies when matrix element value lands.

z_source canonicalization (Cal #3 pre-stage): Shilov boundary general point z_0. Elie pinned in Step 3+4.

8. Cal cold-read absorptions (Sunday afternoon)

Cal Lane E cross-check (absorbed in-place Lane E doc + acknowledged): m_W/m_Z = $\sqrt{(\text{g/N\_c}^2)}$ is arithmetically identical to P1 §7’s $\sqrt{\text{g/N\_c}}$ = $\sqrt{7/3}$. Lane E’s substantive content is the V_(1,1) adjoint decomposition MECHANISM CANDIDATE for the existing P1 §7 prediction, NOT a new arithmetic anchor. Mechanism-vs-post-hoc-match is the cold-read content pending Cal #187.

Cal #35-candidate firing on Lane C (absorbed in-place Lane C doc + acknowledged): g + rank = N_c^2 substrate-algebraic identity appears in THREE contexts (Weinberg + bulk-color + m_W/m_Z reframing). Per Grace + Elie + Keeper synthesis: “1 identity reused, NOT 3 events.” Independence-vs-shared question is Cal #188 cold-read content; Lane C cross-link claim HELD pending Cal verification.

Cal #185 PASS on P1 v0.7 (recorded); 2 non-blocking follow-ons absorbed in-place pre-dispatch.

9. Honest tier disposition (Two-Tier + Cal #182 + Calibration #35-candidate)

RIGOROUS (this v0.2 prep): - Hardy decomposition (Knapp-Wallach 1976 + FK 1994 Ch. XII-XIII). - Bergman canonical metric is Einstein on D_IV^5 (Helgason 1962 + Wolf 1972 standard theorem). - ρ_commit(τ) = heat semigroup on ⍰; standard operator theory. - SO_0(5,2) = automorphism of D_IV^5 = conformal group of 4D Minkowski. - Bergman-Fourier duality between K-types + coherent states.

CANDIDATE (v0.2 prep load-bearing claims): - Substrate-native field equation lives on Shilov boundary; heat + wave = real + imaginary continuations. - ρ_commit = irreversible step of SWPP; emission = unitary U(t) reversible. - Substrate time τ IS the heat-semigroup parameter; arrow of time = positivity of spec(H_B). - Substrate spacetime metric IS the Bergman canonical metric on D_IV^5. - G is the dimensional conversion factor between

$\varkappa$_Bergman and Einstein-equation curvature. - Mass-from-C_2 via per-sector / vacuum subtraction (L4 v0.2 R2-refined).

FRAMEWORK (multi-week): - Black-hole-as-Shilov-saturation quantitative test (K200 G3). - Big-Bang-as-first-commitment commitment (K200 G4 reading-(i)). - G_substrate SI-unit prediction (K200 G5 + Elie Mehler-Helgason). - Λ from boundary equilibrium accumulation rate. - Wick rotation $\tau \leftrightarrow$ it for substrate $\leftrightarrow$ scattering observables (F1). - Mass-from-C_2 explicit derivation for T190 etc. (Lane D + Elie F2).

OPEN (genuinely): - $\hbar$_BST = $\hbar$_Planck identification (Keeper K3). - Operator-density connection ρ_commit(τ) $\leftrightarrow$ ρ_commit(x, t) (Keeper K5). - Lane C bulk-color g+rank=N_c^2 independence-vs-shared (Cal #188). - Lane E mechanism-vs-post-hoc for m_W/m_Z (Cal #187). - 3-generation forcing via heat-kernel framework (Tier 0 v0.1 B1; #414 deep gate). - 3-color forcing via Hardy bulk-color v0.7 (B2).

10. Routing

→ Casey: v0.2 prep draft assembled; Lyra-side material ready for Session 2 joint absorption. K200 gates G1-G5 explicit; Cal cold-read absorptions integrated. G chain Step 1 OPERATIONAL (Elie Toy 3659); Steps 2-4 multi-week paths clear. Standing for your Session 2 trigger.

→ Keeper: Session 2 priority queue per your synthesis: sphere reframe absorption + topology dual-bases + native field equation rewrite + commitment operator consolidation + Cal cold-read absorption + K200 gates G1-G5 ratification. K1-K5 your lane (pin H_B, derive C_2(λ) explicit lowest 20, pin $\hbar$_BST, verify SWPP cycle, connect operator ↔ density). HSL v0.9 + CLAUDE.md headline integration when v0.2 finalizes.

→ Cal: cold-read sequence per your Sunday plan (#186 Lane D → #187 Lane E → #188 Lane C → Tier 0 v0.1.6 with K200 gates). v0.2 prep ready for cold-read after Session 2 joint consolidation; specific gates G2/G3/G4 absorbed per K200 pre-stage.

→ Elie: G chain Step 2 (Helgason $\varkappa$_Bergman closed-form, Toy 3660) is multi-week target per Casey directive. Step 1 OPERATIONAL via your Toy 3659; Step 2 explicit derivation is load-bearing pull. Standing for your Toy 3660 progress.

→ Grace: v0.2 prep available for catalog cross-reference (V_(1,1) decomposition + V_(1/2,1/2) + V_(0,0) vacuum + Shilov boundary structure + Bergman canonical metric). Catalog at CANDIDATE per Cal #182.

→ me: continuing pulls per Casey directive — Dictionary expansion 5 → 10 next + Bulk-color v0.7 mechanism deepening after.

— Lyra, Tier 0 v0.2 LYRA PREP DRAFT (Session 2 with Keeper consolidation). Consolidates v0.1 + v0.1.5 + v0.1.6 + G_substrate sketch into single coherent ~10-section draft; K200 gates G1-G5 explicit; Cal cold-read absorptions integrated; Cal #185 PASS on P1 v0.7 recorded; G chain Step 1 OPERATIONAL per Elie Toy 3659. Awaiting Keeper Session 2 joint absorption + K1-K5 fill + audit ratification. Per Casey pull-continuously directive: this draft is the consolidation work pull, not the Session 2 trigger; Keeper engages when Casey signals.